

The Link Function Problem in Psychological Interaction Testing

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Abstract

Interaction tests are a routine part of psychological analysis, but an interaction claim is not defined by a product term alone. It also depends on the scale on which additivity is assumed. This scale problem is not new, but we reframe it in the modern GLM language of outcome family and link function. We first report a preregistered descriptive review of recent psychological articles, showing that interaction tests are common and that family, link, and scale-of-additivity choices are often implicit, especially for bounded, discrete, ordinal, and sum-score outcomes. We then develop a link-centered framework for distinguishing wrong-family, wrong-link, and wrong-metric problems, and for separating interactions on the linear predictor scale from interactions on the observed response scale. Worked examples focus on forced-choice accuracy with a non-zero chance floor, sum scores as bounded and discrete outcomes, and within-family link choices such as logit versus probit. Simulations show how inadequate links can generate pseudo-interactions, while diagnostic checks may have limited power to identify the wrong link before the interaction becomes statistically detectable. We do not argue that all interactions are artifacts. We argue that many interaction claims are underspecified unless researchers state, justify, and probe the scale on which no interaction is assumed.

Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

The Link Function Problem in Psychological Interaction Testing

Psychological theories often ask whether the effect of one variable depends on another. Stress may impair performance more strongly among people with high anxiety. A treatment may help some participants more than others. A developmental difference may be large at one age and small at another. These are interaction claims: claims about moderation, boundary conditions, and heterogeneous effects. In practice, testing such claims has become part of the standard analytic toolkit in psychology, sometimes almost a routine step after main effects have been examined. This routine use is understandable, because interactions often express important theoretical ideas. Yet an interaction claim should not be interpreted automatically from the presence or absence of a product term ([Rimpler et al., 2026](#); [Rohrer & Arslan, 2021](#)).

Statistically, many interaction claims can be understood as claims about a difference between differences. Does the treatment-control difference change across groups? Does the age-related difference differ between populations? Does the effect of task difficulty vary as anxiety increases? This formulation is intuitive, but it hides a crucial assumption. A difference is always defined on some scale. An interaction is a difference between differences, but differences are not scale-free.

The scale problem has been recognized for a long time. Loftus argued that interactions involving response probabilities are meaningful only relative to an assumed mapping between the observed response and the theoretical component of interest ([Loftus, 1978](#)). Cox treated interaction as a statistical concept tied to model formulation, transformation, and interpretation ([Cox, 1984](#)). Wagenmakers and colleagues later emphasized the distinction between removable and nonremovable interactions, showing that some interactions depend on the measurement scale and therefore do not support unambiguous conclusions about latent psychological processes ([Wagenmakers et al., 2012](#)). Lubinski and Humphreys illustrated how apparently substantive moderator effects can arise from model specification choices rather than genuine psychological synergy ([Lubinski & Humphreys, 1990](#)). More recently, Rohrer and Arslan argued that interaction tests can provide precise answers to vague questions when researchers do not specify the

estimand, scale, or substantive meaning of the interaction (Rohrer & Arslan, 2021).

A related literature has sharpened the problem for modern psychological data. Domingue and colleagues showed that misspecifying the outcome model can produce bias and false discoveries in interaction analyses, especially when outcomes are binary, count, censored, or noninterval (Domingue et al., 2024). McCabe and colleagues showed that, in generalized linear models (GLMs) for probabilities and counts, product-term coefficients are not generally equal to interaction effects on the response scale (McCabe et al., 2022). Work on nonlinear probability models reaches the same practical conclusion: interaction effects should often be evaluated as conditional marginal effects or response-scale contrasts, not as product coefficients alone (Ai & Norton, 2003; Mize, 2019). More broadly, estimand-centered approaches recommend treating models as tools for generating substantively meaningful quantities rather than interpreting coefficients mechanically (Rohrer & Arel-Bundock, 2026). Rimpler and colleagues further emphasized that a theoretical claim of conditionality does not automatically imply that the correct statistical representation is a multiplicative product term (Rimpler et al., 2026). Together, these contributions show that interaction testing is not only a matter of adding the right-hand-side product term to a regression equation.

This paper focuses on one specific part of this broader problem: the role of the link function in GLM interaction testing. In the standard GLM architecture, the family and the link answer two different modeling questions (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972). The family answers: what kind of outcome am I modeling? It specifies the support and sampling process of the response, for example binary choices, counts, positive continuous times, or approximately Gaussian scores. The link answers: on which scale do I assume that effects add? It maps the conditional mean of the outcome to the linear predictor and therefore defines the scale on which a model without a product term is additive.

Choosing an appropriate family is therefore necessary, but not sufficient, for interaction testing. Two models can respect the same outcome family while defining different no-interaction baselines. For response times, for example, a Gamma family may be plausible because the

outcome is positive, continuous, and right-skewed. Yet `Gamma(link = "log")` assumes additivity in log mean response times, which corresponds to proportional change on the millisecond scale, whereas `Gamma(link = "identity")` assumes additivity directly in milliseconds. Neither choice is automatically correct. The relevant question is whether the scientific claim concerns proportional change, absolute change, or some other scale. Link misspecification can affect likelihood-based inference in GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The same logic applies to accuracy data, where the issue can be even more visible. A binomial family may correctly represent the sampling process of correct and incorrect responses, but the link still defines the scale of additivity. A standard logit link maps the linear predictor to the full 0-to-1 probability range. In a two-alternative forced-choice task, however, the theoretically meaningful lower bound may be chance performance, that is, .50. A chance-corrected link can encode this lower asymptote directly. Thus, the distinction is not only between linear models and GLMs. It is also between plausible links within a broadly appropriate family.

Figure 1 shows why link choice is a scale choice, not only a curve-fitting choice. In each panel, the grey construction lines are anchored to a target link: the chance-corrected logit for forced-choice accuracy and the log link for Gamma mean response times. Equal steps on that target-link scale map to unequal steps on the observed response scale. Conversely, the same observed response values would not be equally spaced on an alternative link scale. This is the central reason why link choice matters for interactions: a model without a product term is additive only on the scale defined by its link. When another plausible link is used, the same response-scale pattern can imply a different no-interaction baseline and therefore a different moderation claim.

Figure 2 uses one simulated two-alternative forced-choice accuracy dataset as a motivating example. Each participant completes 50 trials, and the outcome is the proportion correct. The predictors are age and group. The data-generating model includes an age effect and a group effect, but no age-by-group term on the chance-corrected logit scale, the scale on which the task has a .50 chance floor. The point of the figure is not that one can obtain any conclusion by changing links.

The point is narrower: the same data can support different apparent moderation claims when models define different no-interaction baselines (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022).

This does not mean that all interactions are artifacts, or that scale choices make interaction claims arbitrary. Some interactions will be robust across plausible links and response-scale summaries. Others will not. The problem is that many interaction claims are underspecified unless researchers state which scale defines additivity, whether the reported interaction is on the linear predictor scale or the observed response scale, and whether alternative plausible links lead to the same substantive conclusion. In this sense, link choice is not a technical afterthought. It is part of the scientific claim.

The link-function problem should also be separated from two related problems. A wrong-family problem occurs when the model does not respect the outcome type, for example when counts or proportions are treated as Gaussian identity outcomes without justification. A wrong-link problem occurs when the family is broadly plausible, but the link defines an inadequate scale of additivity, for example when a standard binomial logit link is used for a forced-choice task with a non-zero chance floor. A wrong-metric problem occurs when the observed score is treated as if it were already an interval-scale measure of the target construct, for example when a Likert item or sum score is interpreted without considering its ordinal, bounded, or nonlinear relation to the latent quantity. These problems can overlap, but they are not the same.

Our contribution is therefore narrow. We do not claim that link functions have been ignored, nor that scale dependence is a new idea. Instead, we argue that psychology lacks an explicit, interaction-focused framework for choosing among plausible link functions and for quantifying how wrong or inadequate link choices can generate pseudo-interactions in realistic psychological settings. This framework is especially important when researchers have already moved beyond a simple linear model and chosen a broadly plausible GLM family, but have not examined whether the link function encodes the scale of additivity implied by their scientific question.

We develop this argument in three applied cases. The central case is forced-choice accuracy with a non-zero chance floor. Here, a standard binomial model may be right about the sampling process but wrong about the lower bound. The second case is sum scores, which are common in psychology but often discrete, bounded, skewed, and only indirectly related to latent constructs. We do not argue that sum scores are invalid. We argue that interaction claims based on sum scores require scale-aware sensitivity analyses when additivity on the manifest sum-score scale is not clearly justified, especially because ordinal and metric analyses can diverge in ways that affect interactions ([Liddell & Kruschke, 2018](#)). The third case is within-family link choice, such as logit versus probit. These links are often practically similar in the middle of the probability range, but their differences can matter for interaction claims near floors, ceilings, chance levels, or distributional tails.

We also discuss diagnostics, but with a limited claim. Link tests, residual diagnostics, calibration checks, and predictive checks can reveal important forms of misfit ([Dunn & Smyth, 1996](#); [Hartig, 2024](#); [Pregibon, 1980](#)). They should be part of responsible modeling practice. However, diagnostics do not by themselves certify the scientific validity of an interaction claim, and they may not distinguish a wrong link from a missing nonlinearity, a missing interaction, a wrong family, or a problematic measurement metric. For this reason, we treat diagnostics empirically in one focused worked example, asking whether the wrong link is detected in the same conditions in which it can generate a false-positive interaction.

The rest of the paper proceeds as follows. First, we report a preregistered descriptive review of current practice in psychological interaction testing. This review asks whether the issue is only a technical corner case or whether it matters for current practice. Second, we define the link-function problem and distinguish interactions on the linear predictor scale from interactions on the observed response scale. Third, we examine forced-choice accuracy, sum scores, and within-family link choice as applied cases where link assumptions can matter for moderation claims. Fourth, we use simulations to quantify when wrong or inadequate link choices generate pseudo-interactions. Fifth, we evaluate diagnostics in one selected scenario to show what they can

and cannot detect. We close with a minimal workflow for link-aware interaction testing in psychological research.

Current Practice in Psychological Interaction Testing: A Preregistered Review

Why This Review Matters

Is this a technical corner case, or does it matter for current psychological practice? This question motivated our preregistered descriptive review. The review was designed to establish whether the link-function problem is relevant for the kinds of interaction analyses that psychologists actually report. If interaction tests were rare, or if researchers routinely stated the outcome family, link function, and scale of additivity, the problem would be less urgent. If, instead, interaction tests are common and these modeling choices are often implicit, then many moderation claims are difficult to evaluate because the no-interaction baseline is not fully specified.

The review should not be read as an audit of whether individual articles were wrong. In many cases, published reports do not provide enough information to reconstruct the authors' full modeling rationale. A paper may report a model family but not the link, analyze a transformed or aggregated outcome without stating the implied measurement scale, or target an observed-scale estimand that could be defensible but is not described in those terms. For the same reason, we did not systematically code whether reported interactions were removable or nonremovable. That distinction is central to the classical scale-dependence literature ([Cox, 1984](#); [Loftus, 1978](#); [Wagenmakers et al., 2012](#)), but classifying it reliably from published reports alone would often require information about the raw outcome, the assumed measurement scale, the link or transformation, and the substantive scale of interest. Our coding choice was therefore conservative. We asked what could be learned from reported practice: whether interaction tests are frequent, whether link functions are explicit, and whether many analyses involve outcomes for which identity-link additivity is not self-evident.

This makes the review a substantive entry point for the rest of the paper. The main issue is not that all reported interactions are artifacts. The issue is that many interaction claims are hard to

interpret when the family, link, and scale of additivity remain implicit. This focus on transparent analytic choices is consistent with broader recommendations that quantitative manuscripts should report modeling decisions clearly enough for readers to evaluate them (Mustillo et al., 2018).

Review Method

The review protocol was preregistered before the current manuscript was written. It specified a descriptive review of recently published psychological research articles, with four primary goals: estimating how often empirical articles test interactions, how often interaction-testing articles use non-identity links, how often the link function is explicitly reported, and which response-variable types are used in interaction analyses. The protocol also stated that the review was not intended to be an exhaustive systematic review of psychology as a whole, but a targeted estimate of current practice in selected journals.

Articles were sampled from five journals selected to cover different areas of psychology: *Psychological Science*, *Journal of Experimental Psychology: General*, *Journal of Personality and Social Psychology*, *Developmental Science*, and *Psychological Medicine*. Eligible articles were empirical articles presenting original psychological research. Reviews, meta-analyses, editorials, theoretical articles without empirical data, qualitative-only articles, replication studies, and psychometric validation studies were excluded. Articles were then coded in detail if they tested at least one interaction effect, either through statistical modeling or analysis of variance.

For each interaction-testing article, coders recorded whether at least one tested interaction used a non-identity link function, whether the link function was explicitly reported, whether at least one interaction was analyzed with an identity link when another link appeared more appropriate given the outcome, whether at least one such interaction was statistically significant or interpreted analogously, and which response-variable types were analyzed. The protocol specified descriptive summaries, including proportions and Wilson 95% confidence intervals for Boolean variables. The full review protocol, coding definitions, and extended descriptive tables are reported in the Supplement.

Review Findings

The review identified 331 eligible empirical articles. Interaction testing was common: 200 of these articles tested at least one interaction, corresponding to 60.4%, 95% CI [55.1, 65.5]. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in most interaction-testing articles, the link function was either identity by default, not reported explicitly, or not foregrounded as part of the interaction claim.

A large proportion of interaction-testing articles involved outcomes for which identity-link additivity was not obviously innocuous. Among the 200 articles testing interactions, 144 were coded as including at least one potentially problematic identity-link analysis, 72%, 95% CI [65.4, 77.8]. This code should be interpreted cautiously. It means that, given the reported outcome and model, another link appeared plausibly more appropriate. It does not prove that the specific interaction claim was false. Among these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. Table 1 summarizes these descriptive results.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (86 articles), sum scores or composites (77 articles), ordinal, Likert, or rating outcomes (49 articles), response times or durations (29 articles), and counts or error counts (12 articles). This heterogeneity matters because the identity link has different implications across outcome types. For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

What the Review Shows, and What It Does Not Show

The review shows that interaction testing is a routine part of current psychological practice, and that many interaction claims are made in settings where the scale of additivity is not trivial. It also shows that link functions are rarely made explicit. This combination is the key empirical result. If the link function defines the scale on which effects are assumed to add, then an implicit link function implies an implicit no-interaction baseline.

The review does not show that the coded articles reached wrong conclusions. Some identity-link analyses may have been reasonable approximations. Some interaction claims may be robust across plausible links. Some authors may have intentionally targeted observed-scale differences, even if this was not stated in link-function language. The review also does not classify individual interactions as removable or nonremovable, because doing so would often require stronger information than the published report provides. The descriptive result is more modest, but still important: many published interaction claims are underspecified from the reader's perspective because the family, link, and scale of additivity are not made sufficiently explicit.

The next section develops the link-function problem directly. We first distinguish the outcome family from the link function, then show why choosing the family is not enough for interaction testing, and finally separate interactions on the linear-predictor scale from interactions on the observed response scale.

The Link Function Problem

The review shows that interaction tests are common, while the family, link, and scale of additivity are often left implicit. This section explains why that matters. In a generalized linear model (GLM), the expected outcome for observation i , denoted $\mu_i = E(Y_i)$, is related to a linear predictor through a link function (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972):

$$g(\mu_i) = \eta_i.$$

With two predictors, X_i and Z_i , and their product term, the linear predictor is usually written as

$$\eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

This notation separates two decisions that are often blurred. The family answers the question: what kind of outcome am I modeling? It specifies the support and sampling process of the response, such as Gaussian scores, binomial successes, Poisson counts, or positive continuous outcomes. The link answers a different question: on which scale do I assume effects add? It maps the conditional mean of the outcome to the linear predictor. Thus, choosing the family is necessary but not sufficient. A plausible family can keep predictions in the right range, but the link still defines the scale on which a model without a product term is additive.

i What the link function does

The linear predictor, η , is the model's additive scale. Intercepts, main effects, product terms, covariates, and random effects are combined on this scale. Many psychological outcomes live on a different scale. Counts cannot be negative. Probabilities must lie between 0 and 1. Forced-choice accuracy may have a theoretical floor above 0, for example .50 in a true/false task. Response times are positive and often right-skewed. Sum scores have lower and upper bounds, and are often discrete.

The inverse link, $g^{-1}(\eta)$, maps the additive model back to the expected response. This mapping is not neutral for interactions. Equal changes in η can become unequal changes in probabilities, counts, milliseconds, or score units. Conversely, equal observed-scale differences need not be equal differences on the link scale. The link function therefore defines what “no interaction” means in the fitted model.

The distinction is hidden in ordinary linear models because the identity link makes the two scales coincide. If $g(\mu) = \mu$, then additivity on the linear predictor scale is also additivity on the expected response scale. For nonlinear links, this equivalence disappears. A logit link defines additivity in log odds. A probit link defines additivity on a latent-normal scale. A log link defines additivity in log means, which corresponds to multiplicative change on the response scale. The

same general point applies within a family. As noted in the introduction, a Gamma model for response times may use an identity, log, or inverse link. The family may be plausible in all cases, but the implied no-interaction baseline is different.

A simple count example

Suppose children make fewer errors as they get older. A Poisson model with a log link can represent this as a linear age effect on the log mean,

$$\log\{E(Y)\} = \beta_0 + \beta_1 \text{age},$$

with $\beta_1 < 0$. On the observed count scale, the same model implies $E(Y) = \exp(\beta_0 + \beta_1 \text{age})$, a curved decreasing function. The curvature is not an extra psychological mechanism. It is the expected consequence of mapping an additive model on the log scale back to non-negative counts.

Now add a group difference in overall error rate, but no age-by-group product term on the log scale. On the observed count scale, the absolute age-related reduction will usually be larger where expected errors are high and smaller where expected errors are low. A Gaussian identity model fitted to raw counts may treat this changing absolute difference as moderation. A Poisson-log model can describe the same pattern as additive main effects on the log-count scale. The point is not that Poisson-log is always correct. The point is that product-term coefficients and response-scale contrasts answer different questions for counts and other non-Gaussian outcomes ([Domingue et al., 2024](#); [McCabe et al., 2022](#)).

The same logic applies directly to interaction coefficients. The coefficient β_3 is a product-term interaction on the link scale, $g(\mu)$. When $\beta_3 = 0$, the fitted model assumes additivity on that scale. It does not necessarily assume additivity on the observed response scale. A response-scale interaction can instead be defined as a difference between expected differences. For two binary predictors, one such quantity is

$$\{E(Y \mid X = 1, Z = 1) - E(Y \mid X = 0, Z = 1)\} - \{E(Y \mid X = 1, Z = 0) - E(Y \mid X = 0, Z = 0)\}.$$

In a nonlinear GLM, this quantity depends on the inverse link, $g^{-1}(\eta)$. It can be nonzero when the product-term coefficient is zero, and it can differ in size or sign from the product-term coefficient when the product term is nonzero. This is why product-term coefficients in GLMs should not be treated as automatic answers to response-scale moderation questions (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

The link-function problem is therefore a scale-of-additivity problem. To ask whether X and Z interact is to ask whether their effects depart from additivity on some scale. An identity-link model asks whether they depart from additivity on the expected observed mean. A binomial-logit model asks whether they depart from additivity in log odds. A binomial-probit model asks whether they depart from additivity on a probit scale. A chance-corrected binomial link asks whether they depart from additivity on a scale that maps the linear predictor onto the interval from chance performance to perfect performance. These are related statistical models, but they do not express the same scientific null.

This point helps separate three problems that are often conflated.

The present paper focuses mainly on the second problem, while keeping the other two visible. This focus is narrower than the general warning that linear models can be inappropriate for non-Gaussian outcomes. Researchers may choose a broadly appropriate family and still leave the scientific scale of additivity unspecified. For instance, a binomial family may correctly represent successes out of trials, but it does not choose among logit, probit, complementary log-log, flexible generalized links, or a chance-corrected link with a lower asymptote (Stukel, 1988). Link misspecification can affect inference in binary GLMs and GLMMs, and diagnostic evidence may not identify a unique correct link (Czado & Santner, 1992; Yu & Huang, 2019).

A link-aware interaction analysis therefore begins with four questions. First, what is the estimand: a link-scale coefficient, a probability difference, a marginal effect, a risk ratio, a count difference, or another response-scale contrast? Second, what constraints does the outcome impose,

such as bounds, discreteness, ordinal categories, skew, ceilings, floors, or chance performance? Third, which scale of additivity is scientifically plausible for the process under study? Fourth, does the substantive conclusion survive plausible alternative links? These questions do not make interaction testing automatic, but they make the assumptions visible enough to evaluate.

This framework also clarifies the role of the simulations that follow. The simulations do not ask whether every interaction in psychology is an artifact. They ask whether an interaction can be generated, removed, or distorted when the fitted model assumes additivity on a different scale from the data-generating process. This matters because many psychological outcomes have exactly the properties that make link assumptions consequential: bounded accuracy, non-zero chance levels, ordinal ratings, counts, response times, and sum scores. The next sections develop these cases in turn.

Box 1. Prediction, Explanation, and the Scale of Interaction

The importance of link choice depends on the goal of the analysis. For purely predictive purposes, the primary criteria may be predictive accuracy, calibration, generalizability, and decision usefulness. A model can be useful if it predicts future observations well, even if its coefficients do not provide a faithful description of the psychological process that generated the data.

Scientific interaction claims require a different standard. When researchers claim that an effect is moderated, they usually claim more than that one model predicts slightly better than another. They claim that the effect of one variable changes across levels of another variable in a way that is meaningful for theory. That claim requires an estimand. It also requires a scale. A treatment effect might be larger in absolute probability points, larger in odds ratios, larger on a latent response scale, or larger only after a transformation. These are different claims, and they can lead to different conclusions.

This distinction is consistent with estimand-centered workflows that treat models as tools for generating clear target quantities rather than as collections of coefficients to be interpreted mechanically ([Rohrer & Arel-Bundock, 2026](#)). For interaction testing, the relevant target quantity

should be stated before the product term is interpreted. If the target is a response-scale difference-in-differences, the analyst should compute and report that quantity. If the target is additivity on a latent or link scale, the link function should be justified as part of the scientific model.

The point is not that prediction and explanation are separate in practice. Predictive checks, calibration, and out-of-sample performance can all inform scientific modeling. The point is that good prediction alone does not specify the scale on which a moderation claim is being made. For scientific inference about interactions, the link function is part of the claim.

Forced-Choice Accuracy and the Non-Zero Chance Floor

Forced-choice accuracy is a useful case because it separates the outcome family from the link scale. If a participant completes k_i trials and answers Y_i trials correctly, the binomial sampling model is often the natural starting point:

$$Y_i \sim \text{Binomial}(k_i, p_i),$$

where p_i is the expected probability of a correct response. This choice concerns the sampling process. It does not, by itself, decide the lower bound that is meaningful for the psychological task. In a 2-alternative forced-choice task, or in a true/false task when random responding between two alternatives is the relevant chance process, guessing implies an expected accuracy of .50. More generally, in an m -alternative forced-choice task, the chance level is $c = 1/m$. For these tasks, the theoretically relevant response range is often not 0 to 1, but chance to 1.

This distinction gives three increasingly specific specifications. The weakest is a Gaussian identity model fitted to proportions. It ignores the binomial trial process and treats accuracy as an approximately continuous outcome. It also assumes additivity directly in percentage-correct units and can generate fitted values outside the admissible 0-to-1 range. This model may sometimes be useful as a rough descriptive approximation, especially far from the boundaries, but it does not encode either the trial structure or the task-implied lower bound.

A standard binomial logit or probit model improves on this by respecting the binomial sampling process and constraining fitted probabilities to the interval from 0 to 1:

$$p_i = F(\eta_i),$$

where F is the logistic cumulative distribution function for a logit link, or the standard normal cumulative distribution function for a probit link. This model is right about the binomial structure of the data. However, in a forced-choice task with a known chance level, it still defines the lower asymptote as 0. A standard binomial model is right about the sampling process, but wrong about the lower bound.

A chance-corrected binomial link keeps the binomial sampling model but maps the linear predictor to the interval from chance to 1:

$$p_i = c + (1 - c)F(\eta_i).$$

For a 2-AFC or true/false task with chance level .50, this becomes

$$p_i = .50 + .50F(\eta_i).$$

Equivalently, the link is applied to the above-chance quantity $(p_i - c)/(1 - c)$ rather than to the raw probability p_i . The no-interaction baseline is therefore different. A product term in a standard binomial-logit or binomial-probit model tests additivity on a scale defined over the full 0-to-1 probability range. A product term in a chance-corrected binomial model tests additivity on a latent performance-above-chance scale that already encodes the task-implied lower asymptote. These are different modeling claims, even though both use a binomial family.

The difference matters most when one group, condition, or region of the predictor space lies close to chance. Suppose that age has the same effect in two groups on the latent performance-above-chance scale, with no true age-by-group product term on that scale. On the observed accuracy scale, the lower-performing group can still show compression near .50, simply because there is little meaningful room below chance. A standard logit or probit link places this

lower compression near 0 rather than near .50. The fitted model can then use an age-by-group product term to absorb systematic curvature created by the misplaced lower bound. The resulting interaction is not merely random Type I error. It is a pseudo-interaction produced by a mismatch between the scale used by the model and the scale implied by the task. This is consistent with broader work showing that outcome-model misspecification can create false discoveries in interaction analyses ([Domingue et al., 2024](#)), with work showing that product-term coefficients in nonlinear probability models are not automatically response-scale interaction effects ([Ai & Norton, 2003](#); [McCabe et al., 2022](#); [Mize, 2019](#)), and with evidence that binary-link misspecification can affect likelihood-based inference ([Czado & Santner, 1992](#); [Yu & Huang, 2019](#)).

This does not mean that every standard binomial model for accuracy is wrong. The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. Other cases may require richer models, for example if there are lapses, response biases, item heterogeneity, learning across trials, systematic below-chance performance, or participant-specific guessing strategies. The point is not to replace every standard binomial model with a chance-corrected model. The point is to ask whether the link used for the interaction test encodes the scale on which no interaction is scientifically meaningful.

The same dataset can also be analyzed for different purposes. If the goal is prediction, standard and chance-corrected binomial models can be compared by predictive accuracy, calibration, and generalizability. If the substantive estimand is an observed-scale quantity, such as a difference in percentage correct or a difference between such differences, that quantity should be reported directly as a response-scale contrast. If the claim concerns latent performance above chance, ability, sensitivity, or propensity to answer correctly beyond guessing, a chance-corrected link is often closer to the scientific claim. Thus, for interaction claims, the key question is not only whether accuracy is binomial. It is whether the model defines additivity on the scale required by the claim ([Loftus, 1978](#); [McCabe et al., 2022](#)).

Sum Scores as Bounded and Discrete Outcomes

Sum scores make the link-function problem inseparable from measurement. Many psychological outcomes are reported as totals or averages across questionnaire items, task items, symptoms, errors, or ratings. These scores are often treated as if they were direct observations of an approximately continuous psychological dimension. But the latent trait and the observed score are different objects. The latent dimension may be approximately continuous and roughly normal, whereas the observed score is produced by a finite set of item responses through thresholds, endorsement probabilities, success probabilities, or category labels. The mapping from latent trait to observed total is therefore discrete, bounded, and often nonlinear.

This point matters even when the construct itself is well behaved. Even if the underlying psychological dimension were approximately normal, the sum score need not be. A total based on a small number of binary or ordinal items has only a limited number of possible values. It has a lower and upper bound. It can be skewed when items are too easy, too hard, rarely endorsed, or almost always endorsed. It can be compressed near floor or ceiling because there is little room left for the score to move. It can also be non-interval when adjacent raw-score values do not correspond to equal changes in the latent construct. These features are common in psychological variables ([Micceri, 1989](#)), and they are central to why treating ordinal responses as metric can produce false alarms, missed effects, reversals, and misleading interactions ([Liddell & Kruschke, 2018](#)).

This does not mean that sum scores are always wrong or useless. Sum scores can be descriptively useful, transparent, reliable enough for many purposes, and pragmatically tied to how instruments are scored in applied settings. The narrower point is that a Gaussian identity model for a sum score is not neutral. It assumes that effects are additive in raw-score units on the observed score scale. In an interaction model, the no-interaction baseline is that the expected raw-score effect of one predictor is constant across levels of the other predictor. Sometimes this is the target estimand. For example, a clinician, teacher, or researcher may care about a two-point difference on the reported score because decisions and interpretations are made on that metric.

Gaussian identity models are more defensible when the score has many possible values, the distribution is not strongly skewed, observations are not concentrated near floor or ceiling, residual patterns are not strongly tied to fitted values, and the substantive claim is explicitly about raw-score differences. They are less defensible when the score has few levels, is built from ordinal item responses with unclear spacing, shows floor or ceiling concentration, or is interpreted as evidence about moderation on a latent construct scale.

The problem becomes sharper when the verbal claim moves from the manifest score to the underlying construct. A group near the upper bound may show a smaller observed treatment effect not because the treatment is less effective on the construct, but because the score has less room to increase. A group located near a dense region of thresholds may show a larger raw-score change because the instrument is more sensitive there. The product term can then partly reflect the measurement metric rather than genuine moderation in the psychological process. This is the sum-score version of outcome-model and measurement-metric misspecification ([Domingue et al., 2024](#)).

Several pragmatic alternatives can make the scale assumption more explicit. None is automatically correct, and each should be tied to the target claim.

The recommendation is therefore not to abandon sum scores. It is to make the assumed scale visible. Researchers can inspect the score distribution, plot predicted values near the bounds, report response-scale contrasts, and compare Gaussian identity results with ordinal, bounded-count, or item-level alternatives when these are plausible. If the interaction is stable across these specifications, the claim is easier to defend. If it appears only on the manifest sum-score metric, the conclusion can still be informative, but it should be framed as a property of that score, not automatically as moderation of the underlying psychological construct.

Within-Family Link Choice

The link-function problem also arises within the same outcome family. For binary, binomial, ordinal, count, or positive continuous outcomes, the family can be broadly appropriate while the link remains underjustified. This is easy to miss because many default links are

reasonable in many datasets. Logit and probit links, for example, often produce very similar fitted probabilities in the middle of the response range, especially after the usual approximate rescaling of coefficients. That practical similarity is real. It explains why many applied conclusions are unchanged by switching between the two links. However, it is not a general guarantee, especially when the claim depends on interaction, curvature, floor effects, ceiling effects, or tail behavior. Link misspecification can affect likelihood-based inference in binary GLMs and GLMMs ([Czado & Santner, 1992](#); [Yu & Huang, 2019](#)).

The choice between logit and probit is therefore not only a matter of coefficient rescaling. A logit link is natural when the substantive interpretation is in odds or odds ratios. In that case, additivity on the linear predictor scale means additivity in log odds, and exponentiated coefficients have a direct odds-ratio interpretation. A probit link is natural when the response can be represented as threshold crossing on an approximately normal latent variable, such as ability, signal strength, response tendency, propensity, or latent correctness. This does not make probit generally superior to logit. It means that the two links encode different stories about the scale on which effects combine.

For interaction claims, this distinction is more consequential than it may appear from main effects alone. Two links can give nearly overlapping fitted probabilities in the central range but imply different distances on the link scale near the floor, ceiling, chance level, or distributional tails. A product term in a logit model asks whether effects depart from additivity in log odds. A product term in a probit model asks whether they depart from additivity on a latent-normal threshold scale. These null hypotheses are often similar, but they are not identical. Moreover, when the substantive claim concerns observed probabilities, accuracy, endorsement, or risk, the product-term coefficient is not itself the response-scale interaction. The response-scale interaction must be stated as a contrast, marginal effect, or difference between predicted probabilities ([Ai & Norton, 2003](#); [McCabe et al., 2022](#); [Mize, 2019](#)).

This fitted-data example is deliberately simple. Its purpose is not to show that logit and probit usually lead to different conclusions. They often do not. The point is more specific: small

differences in link curvature can be hard to see in the fitted probability curves, yet remain structured rather than random. If a moderation claim depends on the part of the predictor range where this discrepancy is largest, especially near a floor, ceiling, chance level, or tail, link choice can become inferentially consequential.

This is why logit-versus-probit sensitivity should not be treated as a cosmetic robustness check. If a moderation claim appears only under one plausible link, changes sign under another, or depends on predictions near a boundary, the claim should be reported as conditional on the chosen link. Conversely, if the same response-scale conclusion is stable across logit, probit, and other theoretically defensible links, the interaction claim becomes easier to defend. The point is not that logit and probit usually disagree. The point is that an interaction claim is a claim about a no-interaction baseline, and the link defines that baseline.

The same logic applies beyond the logit-probit pair. A binomial family specifies a sampling model for successes out of trials, but it does not decide whether additivity is most plausible on a logit, probit, complementary log-log, flexible generalized-logistic, or chance-corrected scale ([Stukel, 1988](#)). In forced-choice tasks, the link question includes whether the lower asymptote should be 0, as in a standard binomial link, or the theoretical chance level of the task. An ordinal family does not decide which cumulative link is appropriate, and applied ordinal-regression work explicitly treats logit, probit, complementary log-log, negative log-log, and cauchit links as different modeling choices ([Smith et al., 2020](#)). A Gamma family does not decide whether the mean should be additive on the identity, log, or inverse scale. For interaction testing, these choices are not merely alternative parameterizations. They define different scales on which no interaction is assumed.

When link choice may matter less

The link-function problem does not imply that every link choice will change the substantive conclusion. In many applications, plausible links lead to nearly identical fitted values and similar response-scale summaries. The risk is smaller when predicted probabilities remain far from floor and ceiling, effects are modest, the relevant predictor range is well observed, and

candidate links produce almost overlapping curves in that range. As Figure 4 illustrates, the relevant issue is not whether two fitted curves look dramatically different. Often they do not. The issue is whether the pointwise discrepancy between them overlaps with the region that drives the interaction claim. Link choice is also less threatening when the interaction is reported as a response-scale quantity, such as a marginal effect or contrast, and when that quantity is robust across theoretically defensible links (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Some theories also do not depend on tail behavior. If the claim concerns average differences in a central range, rather than saturation, guessing, floors, or ceilings, link sensitivity may be small. This is not a reason to ignore the link. It is a reason to report sensitivity analyses, so that readers can see whether the interaction claim is tied to a narrow modeling convention or survives plausible alternatives.

Diagnostics Are Necessary but Not Sufficient

Diagnostics should be part of link-aware modeling, but they are not a complete solution to the link-function problem. The reason is a diagnostic paradox. A wrong link can induce a systematic pseudo-interaction that becomes statistically detectable as sample size increases, while the diagnostics used to identify the wrong link may still have limited power, or may flag only a broad form of misfit. In that situation, the interaction test can become sensitive to the artifact before the diagnostic evidence becomes specific enough to identify the source of the artifact.

Pregibon-style link tests are one direct response to this problem. They expand the fitted GLM by adding terms derived from the fitted link, then ask whether the expanded model improves the fit (Pregibon, 1980). This is useful because the test targets link adequacy more directly than a generic residual plot. Its limitation is also central. A significant added-term test is a warning that the assumed link may be inadequate, but it does not automatically identify the correct link. The signal may also reflect a missing nonlinear predictor, an omitted covariate, an omitted interaction, or another misspecification of the systematic component. Conversely, a non-significant result does not certify that the link defines the scientifically correct scale of additivity. It only means that this diagnostic, in this sample, did not detect this particular departure.

Simulation-based residual diagnostics provide a broader check. DHARMA implements randomized quantile residuals by simulating new responses from the fitted model and comparing the observed response with the fitted predictive distribution ([Dunn & Smyth, 1996](#); [Hartig, 2024](#)). The resulting residuals are designed to be interpretable across GLMs and GLMMs, where raw, Pearson, and deviance residuals can be difficult to read. In the worked example below, we use four DHARMA checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over age. These checks are useful because they can detect broad misfit, overdispersion or underdispersion, zero-inflation or zero-deflation, and structured residual patterns. Some diagnostics may detect some forms of link-induced misfit, as suggested by dispersion checks in related logit-probit examples. But this depends on the scenario, sample size, outcome distribution, and type of departure. A residual diagnostic that works in one case should not be treated as a general certificate of link adequacy.

Model comparison tools, such as AIC, BIC, WAIC, or LOO, add another source of evidence. They can compare candidate models when the candidate set is meaningful and specified before interpreting the interaction. However, relative fit is not the same as scientific scale validity. Two links can have similar predictive performance while implying different no-interaction baselines, especially near floors, ceilings, chance levels, or distributional tails. Conversely, a model with slightly better predictive fit may not correspond to the estimand that the theory requires. For interaction claims, predictive adequacy and substantive adequacy are related, but they are not identical.

For this reason, diagnostics should be treated as warnings, not certification. They can show that a model deserves scrutiny, and sometimes they can show that a fitted model is plainly inadequate. They usually cannot distinguish, by themselves, between a wrong link, a missing nonlinearity, a missing interaction, a wrong family, and a wrong measurement metric. The practical implication is that diagnostics should be paired with link sensitivity analyses and an explicit definition of the estimand. If the scientific claim concerns response-scale moderation, researchers should report response-scale contrasts or marginal effects and examine whether those

quantities are stable across plausible links ([McCabe et al., 2022](#); [Mize, 2019](#); [Rohrer & Arel-Bundock, 2026](#)). If the claim concerns additivity on a latent or linear-predictor scale, the chosen link needs a theoretical justification. Diagnostics can support that argument, but they cannot replace it.

Simulation Studies

The previous sections established the conceptual problem: the link function defines the scale on which a no-interaction model is additive. The simulations ask how this conceptual problem translates into inferential risk in settings where the true data-generating process is known. Across simulations, the true product-term interaction is set to zero on the generating scale. We then fit models that impose different scales of additivity and record how often they detect a statistically significant interaction. The target is not to estimate how often published interactions are false. It is to quantify when wrong or inadequate links can generate pseudo-interactions under controlled conditions ([Domingue et al., 2024](#); [Rimpler et al., 2026](#)).

i Pseudo-interactions as power against the wrong null

A link-induced false-positive interaction is not only a sampling accident. Under the correctly specified GLM link, the no-interaction null states that effects are additive on the scale defined by that link. Under a wrong link, the same data-generating process may imply a nonzero interaction on the fitted link scale. In that case, a significant product term is better understood as power to detect a pseudo-interaction created by the mismatch between the generating link and the fitted link.

This idea extends the classic scale-dependence argument for interactions ([Cox, 1984](#); [Loftus, 1978](#); [Wagenmakers et al., 2012](#)) and connects it to GLM misspecification. Prior work has shown that outcome-model misspecification can produce biased and false-positive interaction findings ([Domingue et al., 2024](#)), and that product-term coefficients in nonlinear GLMs do not automatically equal interaction effects on the response scale ([Ai & Norton, 2003](#); [McCabe et al., 2022](#); [Mize, 2019](#)). Here, the same logic is used prospectively: if a wrong link induces a

nonzero interaction on the fitted scale, the expected false-interaction rate can be understood as the probability of detecting that induced quantity.

The idea is easiest to see in a simple 2×2 design. Let $\mu_{00}, \mu_{10}, \mu_{01}, \mu_{11}$ be the four expected response means generated from a GLM with no product-term interaction on the target link scale. If the fitted model uses link g , the induced interaction on that fitted scale is

$$\Delta_g = g(\mu_{11}) - g(\mu_{10}) - g(\mu_{01}) + g(\mu_{00}).$$

When g is the target link, $\Delta_g = 0$. When g is not the target link, Δ_g can be different from zero even though the generating model contains no product term. In this 2×2 case, Δ_g is the population link-scale difference-in-differences that the fitted product term is trying to estimate. The probability of a significant interaction is then governed by the size of this pseudo-effect relative to its standard error, approximately $\lambda = \Delta_g / SE(\hat{\Delta}_g)$. This is why the risk depends jointly on the two main effects, the intercept or baseline location, the range of the predictors, the shape of the link, and the precision of the design.

This framing also explains why link misspecification is not equally harmful everywhere. If predicted values remain in a range where plausible links produce nearly the same mean structure, Δ_g may be negligible. Near floors, ceilings, chance levels, or distributional tails, the same main effects can generate a much larger pseudo-interaction. The simulations below estimate this misspecification-induced detection rate in realistic psychological designs.

The box summarizes the logic behind the simulation estimand. The reported detection rates are therefore interpreted as misspecification-induced detection rates, not as estimates of ordinary Type I error under a correctly specified GLM.

The simulations quantify two quantities. First, for each substantive scenario, we estimate the probability of detecting an interaction when the generating model contains no product term on the relevant scale. Second, in a diagnostic worked example, we estimate how often model checks flag the misspecification that produced the pseudo-interaction. This comparison matters because

link-oriented tests and simulation-based residual diagnostics can be useful, but they do not guarantee that the link is adequate for the interaction claim being tested (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

The simulation code is intentionally separated from the manuscript. The main text reports only compact summaries. The full data-generating processes, scripts, and extended output are reported in the Supplement and repository.

Simulation 1: Forced-Choice Accuracy

Simulation 1 quantifies the forced-choice problem under controlled conditions. We generated binomial accuracy data with a .50 chance floor and no age-by-group product term on the chance-corrected logit scale. We then fit four models: a Gaussian identity model for proportions, standard binomial-logit and binomial-probit models, and a chance-corrected binomial-logit model. The outcome of interest was the false-positive rate for the age-by-group product term, that is, how often each fitted model detected an interaction that was absent on the generating scale.

The simulation is designed to be realistic rather than maximal. The sample size and number of trials are moderate, and the scenarios vary the overall performance level so that the wrong-link problem is visible without forcing all misspecified models to fail almost always. The expected pattern is that the chance-corrected model remains close to nominal alpha, because it matches the scale used to generate the data. The Gaussian identity model and the standard binomial links can show elevated false-positive rates, especially when the scenario places one group closer to the chance floor. The important point is not that standard binomial models always produce false interactions. It is that, when the theoretical floor is .50 but the fitted link treats 0 as the lower asymptote, the product term may become a way to repair a wrong no-interaction baseline.

Simulation 2: Sum Scores

Simulation 2 makes the measurement-scale problem concrete. We generated data on a latent continuous scale with a continuous predictor, a group effect, and no true x-by-group product term. We then converted the latent variable into ordinal item responses through item thresholds

and summed the items. The interaction was therefore absent on the latent generating scale, but the observed score was bounded, discrete, and differentially compressed near the floor and ceiling.

The simulation compares three analyses. The observed sum-score identity model is the familiar Gaussian identity analysis of the manifest total. The bounded binomial approximation treats the score as a bounded count, which respects the 0-to-maximum range but remains only an approximation because item responses are ordinal and heterogeneous. The latent generating scale is an idealized benchmark available only in the simulation, because the true latent θ is known. It is included to show what the interaction test would target if the generating construct scale were directly observed.

The expected pattern is not that sum scores always create false interactions. The point is conditional. When the manifest score is approximately linear over the observed range, far from its bounds, and the claim is about raw-score units, the Gaussian identity analysis may be a reasonable approximation. When the score compresses the latent scale unevenly, the raw-score no-interaction baseline can diverge from the latent no-interaction baseline. In that case, a statistically significant product term can reflect the score metric as much as, or more than, moderation of the underlying construct. The practical conclusion is to treat sum-score interactions as scale-specific unless they are stable across plausible measurement and outcome models.

Simulation 3: Within-Family Link Choice

Simulation 3 moves from the fitted-data illustration in Figure 4 to a repeated-sampling question: can within-family link choice inflate false-positive interaction rates when the outcome family, experimental design, and random-effect structure are held fixed? We generated repeated binary responses in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept. The true condition-by-group product term was zero on the generating link scale. We then crossed the generating link and fitted link, using logit and probit in both roles, and fitted all models as random-intercept GLMMs.

This simulation is deliberately narrow. It does not compare a correct binary model with a Gaussian identity model, and it does not change the random-effects structure. The only

misspecification is within-family link choice. The target null is no condition-by-group product term on the generating link scale. Under the wrong fitted link, the same cell probabilities can imply a small nonzero product term on the fitted link scale. With enough information, that deterministic mismatch can become statistically detectable.

The point is not that logit and probit usually lead to different conclusions. They often do not, especially when probabilities are concentrated in the middle of the response range. The point is more specific: when the design places one group or condition closer to a tail, floor, or ceiling, a plausible but wrong link can turn a no-interaction data-generating process on one scale into an apparent interaction on another. Simulation 3 quantifies this risk, whereas Figure 4 shows why the visual similarity of fitted curves is not, by itself, a guarantee of interaction robustness.

The table reports false-positive rates relative to the generating-link null. The median product-term coefficient and its standard error are on the fitted link scale, so they should not be read as response-scale changes in probability. The figure makes the Monte Carlo result explicit: even with the same binary outcome family, logit and probit define slightly different no-interaction baselines, and this difference can become inferentially visible in repeated samples. A result that is stable across both links is easier to defend. A result that appears only under one link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links.

Diagnostic Worked Example

Finally, we examined diagnostics in one selected chance-floor scenario. The data were generated from a chance-corrected binomial model with a .50 lower asymptote and no age-by-group product term on the generating link scale. We then fitted the misspecified standard binomial-logit model, which respects the binomial sampling process but assumes a lower asymptote of 0. The purpose was not to find the best possible diagnostic battery. It was to compare two rates in the same Monte Carlo setting: the false-positive rate for the age-by-group interaction under the wrong link, and the rate at which diagnostic checks flag the same wrong-link problem.

The primary diagnostic workflow used DHARMA residual checks for the fitted standard

binomial-logit model (Dunn & Smyth, 1996; Hartig, 2024). In each replication, we simulated randomized quantile residuals and recorded four p-values: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over age. We report these checks separately. We do not combine them into a single Holm-corrected or uncorrected omnibus decision, because the diagnostics are correlated, partly redundant summaries of the same simulated residuals and have different meanings. Uniformity, dispersion, and residual quantile patterns do not answer the same diagnostic question. A single corrected omnibus decision would therefore be too easy to misread as a general pass-fail test of link adequacy.

We also report a Pregibon-style added-term link check as a secondary comparator (Pregibon, 1980). This check is closer to the link-function problem than a generic residual diagnostic, but it still tests only a restricted alternative. Its role here is not to certify the correct link. Its role is to show whether a link-oriented diagnostic detects the misspecification in the same replications in which the wrong link can produce a false-positive interaction.

The example illustrates the diagnostic paradox. The misspecified standard binomial link can produce a detectable interaction even though the generating model contains no age-by-group product term on the appropriate link scale. Some diagnostics may flag the problem, and this should be valued when it happens. In this scenario, however, the diagnostic signals are not equivalent, and none of them should be interpreted as a definitive test of link correctness. The Pregibon-style check is informative because it is link-oriented, but it still tests a narrow added-term alternative. DHARMA checks are useful because they describe different residual departures, but they do not certify the substantive scale of the interaction. The main lesson is therefore pragmatic: diagnostics are warnings, not certification. For interaction claims, they should be accompanied by a clear estimand, a justified link, and sensitivity analyses over plausible links.

A Minimal Workflow for Link-Aware Interaction Testing

The practical implication is not that researchers should stop testing interactions. It is that interaction claims should be made scale-aware. A minimal workflow is shown in Table 8.

The workflow is not a rule that can make interaction testing automatic. It is a way to put

the main judgments in the open. Researchers still need theory to decide which contrast matters, which scale is meaningful, and which links are plausible. The table simply asks that these decisions be visible before the product term is interpreted. This matters because different quantities can be defensible for different questions: a linear-predictor coefficient may be useful for a model-based claim, whereas a response-scale contrast or marginal effect may be closer to the substantive claim made in words (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Diagnostics and predictive checks can warn about misspecification, but they do not prove that the selected link is the true scientific scale (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980). Sensitivity analyses therefore play a pragmatic role. When the interaction conclusion is stable across plausible links, the claim becomes easier to defend. When it is not stable, the paper can still be informative, but the conclusion should be stated as conditional on a modeling choice. In this sense, link-aware reporting extends general quantitative-reporting principles: choices that affect the claim should be visible, justified, and interpretable (Mustillo et al., 2018).

Discussion

Interactions remain central to psychological theory, and the point of this paper is not that they are usually wrong. Some interaction claims will be robust across reasonable links, scales, and response-scale summaries. Others will not. Our claim is narrower: many interaction claims are underspecified because they do not state the scale on which additivity, and therefore no interaction, is being assumed. In GLMs, this scale is defined by the link function. This makes link choice a substantive modeling decision, not only a technical detail. The same issue may also extend beyond interactions to the broader study of nonlinear effects, because many psychological analyses search for curvature, thresholds, and changing slopes without first specifying the scale on which these patterns are expected to be absent or present (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022; Rohrer & Arslan, 2021; Wagenmakers et al., 2012).

The preregistered review showed why this matters for current practice. Interaction tests were common, but link functions were often implicit, and many interaction-testing articles used identity-link analyses in contexts where the outcome made the scale of additivity worth justifying.

These findings do not show that the reviewed articles reached false conclusions. They show that the scale of many interaction claims is difficult to evaluate from the published report.

The conceptual framework separated three problems that are often mixed together: wrong family, wrong link, and wrong measurement metric. Forced-choice accuracy illustrates the wrong-link problem clearly. A standard binomial family respects trial counts, but a standard link may fail to encode the non-zero chance floor of the task. Sum scores illustrate the measurement-metric problem: an interaction on the manifest score scale may not be an interaction on the latent construct scale. Within-family links illustrate the more general point that plausible links can define different baselines for no interaction.

The simulations were designed to quantify pseudo-interactions, not to estimate the prevalence of false interactions in the literature. They show how link and metric choices can generate significant product terms even when the true interaction is zero on the generating scale. The diagnostic example shows a complementary point: diagnostics are useful, but they do not by themselves certify that a scientific interaction claim is made on the intended scale.

We deliberately reported the DHARMA checks separately rather than collapsing them into a single omnibus decision. In applied work, residual diagnostics are rarely equivalent to one clean hypothesis test. They are better understood as partially redundant probes of different possible symptoms of misspecification. This is precisely why passing one diagnostic, or even several diagnostics, should not be taken as strong evidence that an interaction claim is robust to link choice.

Several limitations are important. First, the review was descriptive and targeted to selected journals, not a census of psychological research. Second, we did not code whether published interactions were removable or nonremovable, because that distinction often cannot be recovered from published reports alone. Third, the simulations are intentionally simple. More realistic data may include random effects, item heterogeneity, lapses, response bias, nonlinear predictors, and missing data. Fourth, chance-corrected links are appropriate only when a chance floor is theoretically justified. They are not a universal replacement for standard binomial models. Finally,

logit and probit links often lead to very similar conclusions, especially away from boundaries; the paper's claim is that link choice can matter for interaction claims, not that it always does.

The main contribution is therefore practical. Link-aware interaction testing does not make moderation claims automatic, and it does not identify one universally correct scale. It asks researchers to make the main judgments visible: what outcome family is plausible, what link defines the no-interaction baseline, what response-scale quantity answers the substantive question, and whether the conclusion survives plausible alternatives. When the same conclusion holds across defensible links, the interaction claim is strengthened. When it changes, the result can still be informative, but it should be reported as conditional on a modeling choice. This workflow complements existing guidance on nonlinear interactions, marginal effects, and prediction-based interpretation (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). In GLMs, the link function is not an afterthought. It is the scale on which no interaction is defined.

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Table 1*Descriptive summary of the preregistered review of interaction testing and link-function reporting.*

Quantity	n	Denominator	Percent	95% CI
Eligible empirical articles reviewed	331	331	100.0%	[98.9%, 100.0%]
Articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using a non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with a potentially problematic identity-link analysis	144	200	72.0%	[65.4%, 77.8%]
Potentially problematic identity-link cases reporting at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denominator	Percent
Binary / accuracy / proportions	86	200	43.0%
Sum scores / composites	77	200	38.5%
Ordinal / Likert / ratings	49	200	24.5%
Response times / durations	29	200	14.5%
Neural / physiological measures	23	200	11.5%
Difference / distance scores	21	200	10.5%
Counts / error counts	12	200	6.0%
Correlations / associations	8	200	4.0%

Table 3*Three distinct sources of scale-related problems for interaction claims.*

Problem	Key question	Psychological example	Risk for interaction claims
Wrong family	Does the model represent the support or sampling process of the outcome?	Counts, proportions, or trial accuracies modeled as unbounded Gaussian observations.	The fitted model may impose impossible predictions or miss the mean-variance relation, so curvature or constraints can appear as moderation.
Correct family, wrong link	Given the outcome family, which scale defines additivity?	Forced-choice accuracy modeled with a standard binomial logit, even though the task has a known chance floor such as .50.	The product term tests additivity on the wrong scale, so a pseudo-interaction can arise even when the correct link-scale model has only main effects.
Wrong measurement metric	Does the recorded score have an interpretable interval scale for the claim being made?	Likert items or sum scores treated as raw interval outcomes without justification.	A product term may reflect thresholds, discreteness, bounds, or scale compression rather than moderation of the underlying construct.

Alternative	Why it helps	When it fails
Cumulative-link or ordered-probit/logit model	It treats the outcome as ordered categories and models threshold crossing rather than equal raw-score distances (Agresti, 2010; Bürkner & Vuorre, 2019; Smith et al., 2020).	It can be awkward for long totals, may rely on proportional-odds or related assumptions, and may still be too coarse if the claim concerns item-level functioning.
Item-level ordinal or IRT-inspired model	It separates the latent construct from item thresholds, item difficulties, and item discrimination, so floor, ceiling, and uneven item sensitivity can be represented directly.	It needs item-level data, stronger measurement assumptions, larger samples, and attention to local dependence, differential item functioning, and model fit.
Binomial, beta-binomial, or Poisson-binomial approximation	It can be useful when the score is a count of successes, symptoms, or endorsements, and it respects discreteness and bounds. Beta-binomial versions can absorb overdispersion, and Poisson-binomial versions can allow heterogeneous item probabilities.	A simple binomial model may wrongly treat items as exchangeable trials. These models may also be poor for ordinal ratings or for items that do not define comparable successes.

Alternative	Why it helps	When it fails
Robust and sensitivity analyses	They show whether the interaction conclusion depends on the manifest sum-score metric, for example by comparing raw-score, ordinal, bounded-count, and item-level analyses.	They do not identify the true scale by themselves. If plausible analyses disagree, the substantive conclusion must be stated as scale-dependent.

Table 4

Simulation 1: false-positive interaction rates and median model-implied observed-scale changes in the group gap for forced-choice accuracy data. CI = Wilson 95% confidence interval.

Scenario	Model	Fits	Sig	False-positive		Median gap change
				rate	95% CI	
Lower performance	Gaussian identity	300	137	0.457	[0.401, 0.513]	-1.779
Lower performance	Standard binomial logit	300	191	0.637	[0.581, 0.689]	-1.769
Lower performance	Standard binomial probit	300	183	0.610	[0.554, 0.663]	-1.781
Lower performance	Chance-corrected binomial logit	178	14	0.079	[0.047, 0.128]	-1.225
Middle performance	Gaussian identity	300	40	0.133	[0.099, 0.176]	-0.865
Middle performance	Standard binomial logit	300	169	0.563	[0.507, 0.618]	-0.862
Middle performance	Standard binomial probit	300	148	0.493	[0.437, 0.550]	-0.866
Middle performance	Chance-corrected binomial logit	300	18	0.060	[0.038, 0.093]	-0.707
Higher performance	Gaussian identity	300	36	0.120	[0.088, 0.162]	0.654
Higher performance	Standard binomial logit	300	103	0.343	[0.292, 0.399]	0.555
Higher performance	Standard binomial probit	300	59	0.197	[0.156, 0.245]	0.617
Higher performance	Chance-corrected binomial logit	300	14	0.047	[0.028, 0.077]	0.526

Table 5

Simulation 2: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x -by-group product term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows the response-scale group gap implied by the same latent group difference; the gap can change across x because the observed score is compressed unevenly. Panel C shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because θ is known in the simulation, not a model available in ordinary applied data analysis.

Scenario	Model	Scale	Fits	Sig	False- positive rate	95% CI	Median gap change
Middle of scale	Gaussian-logit bounded score	sum score	300	13	0.043	[0.025, 0.073]	-0.625
Middle of scale	Latent generating scale	latent θ	300	16	0.053	[0.033, 0.085]	0.018
Middle of scale	Observed sum-score identity	sum score	300	29	0.097	[0.068, 0.135]	-0.547
Near ceiling	Gaussian-logit bounded score	sum score	300	15	0.050	[0.031, 0.081]	1.530
Near ceiling	Latent generating scale	latent θ	300	16	0.053	[0.033, 0.085]	0.003
Near ceiling	Observed sum-score identity	sum score	300	112	0.373	[0.321, 0.429]	1.328
Near floor	Gaussian-logit bounded score	sum score	300	23	0.077	[0.052, 0.112]	-1.916
Near floor	Latent generating scale	latent θ	300	18	0.060	[0.038, 0.093]	0.011
Near floor	Observed sum-score identity	sum score	300	216	0.720	[0.667, 0.768]	-1.762

Table 6

Simulation 3: false-positive interaction rates for the condition-by-group product term in repeated-trial random-intercept GLMMs. CI = Wilson 95% confidence interval.

Generating link	Fitted link	Link status	Fits	Sig	False-positive rate	95% CI	Median beta interaction	Median SE
logit	logit	Matched link	300	16	0.053	[0.033, 0.085]	-0.001	0.075
logit	probit	Wrong link	300	45	0.150	[0.114, 0.195]	-0.041	0.042
probit	logit	Wrong link	300	41	0.137	[0.102, 0.180]	0.071	0.076
probit	probit	Matched link	300	15	0.050	[0.031, 0.081]	-0.003	0.043

Table 7

Diagnostic worked example: false-positive interaction rate and separate diagnostic detection rates in one selected chance-floor scenario. CI = Wilson 95% confidence interval.

Quantity	Fits	Sig	Rate	95% CI
Wrong-link interaction	300	255	0.850	[0.805, 0.886]
DHARMa uniformity	300	1	0.003	[0.001, 0.019]
DHARMa dispersion	300	1	0.003	[0.001, 0.019]
DHARMa residual quantiles over fitted values	300	52	0.173	[0.135, 0.220]
DHARMa residual quantiles over age	300	89	0.297	[0.248, 0.351]
Pregibon-style added-term link check, secondary	300	179	0.597	[0.540, 0.651]

Table 8*A minimal workflow for link-aware interaction testing.*

Step	Question	Why it matters for interactions	Minimal reporting
Define the interaction estimand	What quantity would answer the moderation question?	Product terms, marginal effects, and difference-in-differences can answer different questions.	State the target contrast, scale, moderator values, and whether the estimand is descriptive, associational, or causal.
Inspect outcome constraints	What restrictions does the outcome impose?	Bounds, discreteness, floors, ceilings, and chance levels can make equal observed differences misleading.	Report the response range, discreteness, trial or item count, chance level, and visible floor or ceiling effects.
Justify the outcome family	What sampling model and support are plausible?	A wrong family can create residual structure and pseudo-interactions before link choice is considered.	Name the family and justify it from the outcome type and design, such as trials, counts, positive times, or ordered categories.
Justify the link	On which scale should effects add if there is no interaction?	The link defines the no-interaction baseline. A wrong link can turn curvature into a product term.	Name the link and give the substantive or design reason, such as odds, latent threshold, log mean, or chance floor.
Distinguish linear-predictor interaction from response-scale interaction	Is the claim about the model scale, the observed scale, or both?	A significant product coefficient may not correspond to meaningful response-scale moderation.	Report the product coefficient when relevant, and also predicted means, contrasts, marginal effects, or difference-in-differences.

Figure 1

Same outcome family, different link functions. The grey construction lines are drawn from a target link: the chance-corrected logit in Panel A and the log link in Panel B. Vertical grey lines mark equal intervals on the target-link linear predictor scale. Horizontal grey lines show where those equally spaced target-link values fall on the observed response scale. The key point is that equal intervals on one link scale do not remain equal intervals under another link. Thus, models with the same outcome family can imply different no-interaction baselines, even when their fitted curves look similar over the observed range.

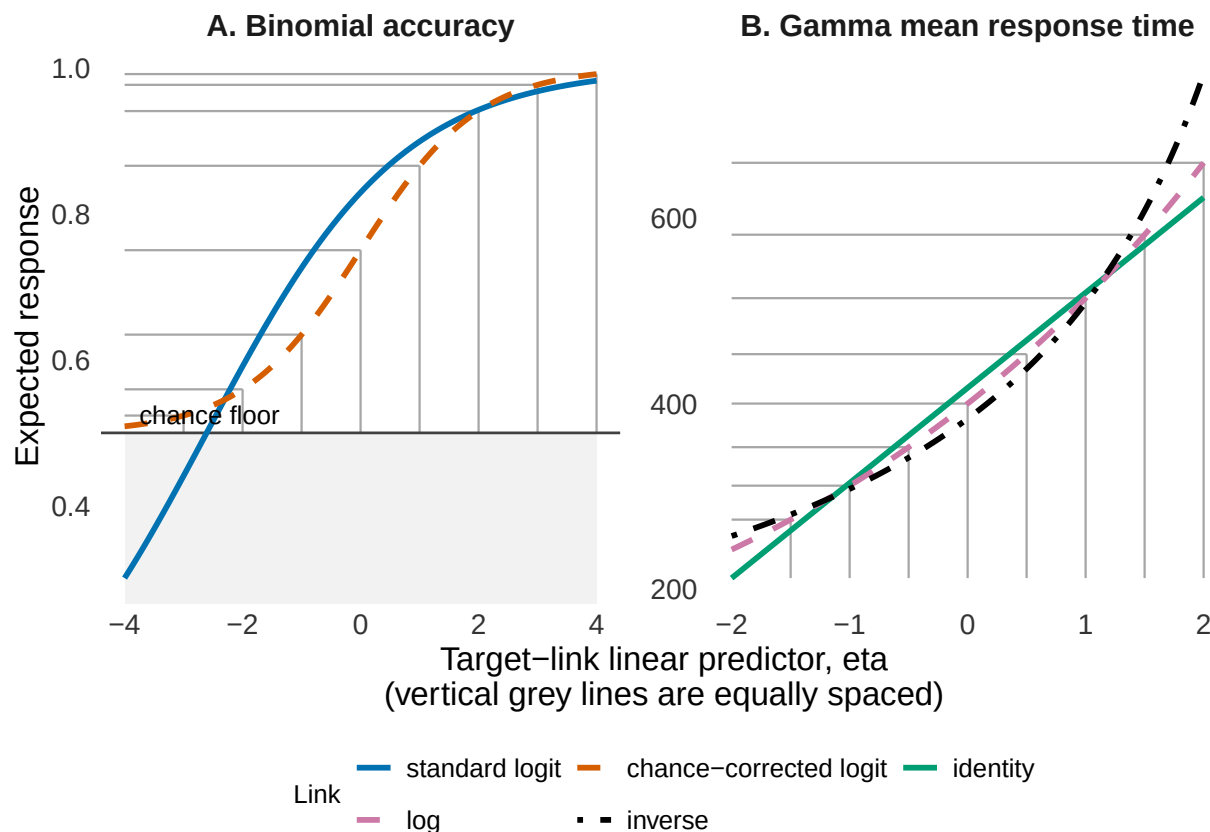


Figure 2

Same data, different no-interaction baselines. Data were simulated from a 2-alternative forced-choice accuracy task with 50 trials per participant and a .50 chance floor. The data-generating model included age and group effects, but no age-by-group interaction on the chance-corrected 2-AFC logit scale. Panel A shows the true response-scale pattern implied by that model. Panel B shows one simulated dataset. Panel C fits the same data with three models. The Gaussian identity model is simple and interpretable on the observed accuracy scale, but it can produce predictions outside the 0-to-1 range and ignores the binomial sampling process. The standard binomial logit/probit model, shown here with a logit link, respects the binomial process and the 0-to-1 range, but assumes a lower asymptote at 0. The chance-corrected binomial link incorporates the task's chance floor. The question is not only which curve fits, but which scale defines the scientific null hypothesis of no moderation.

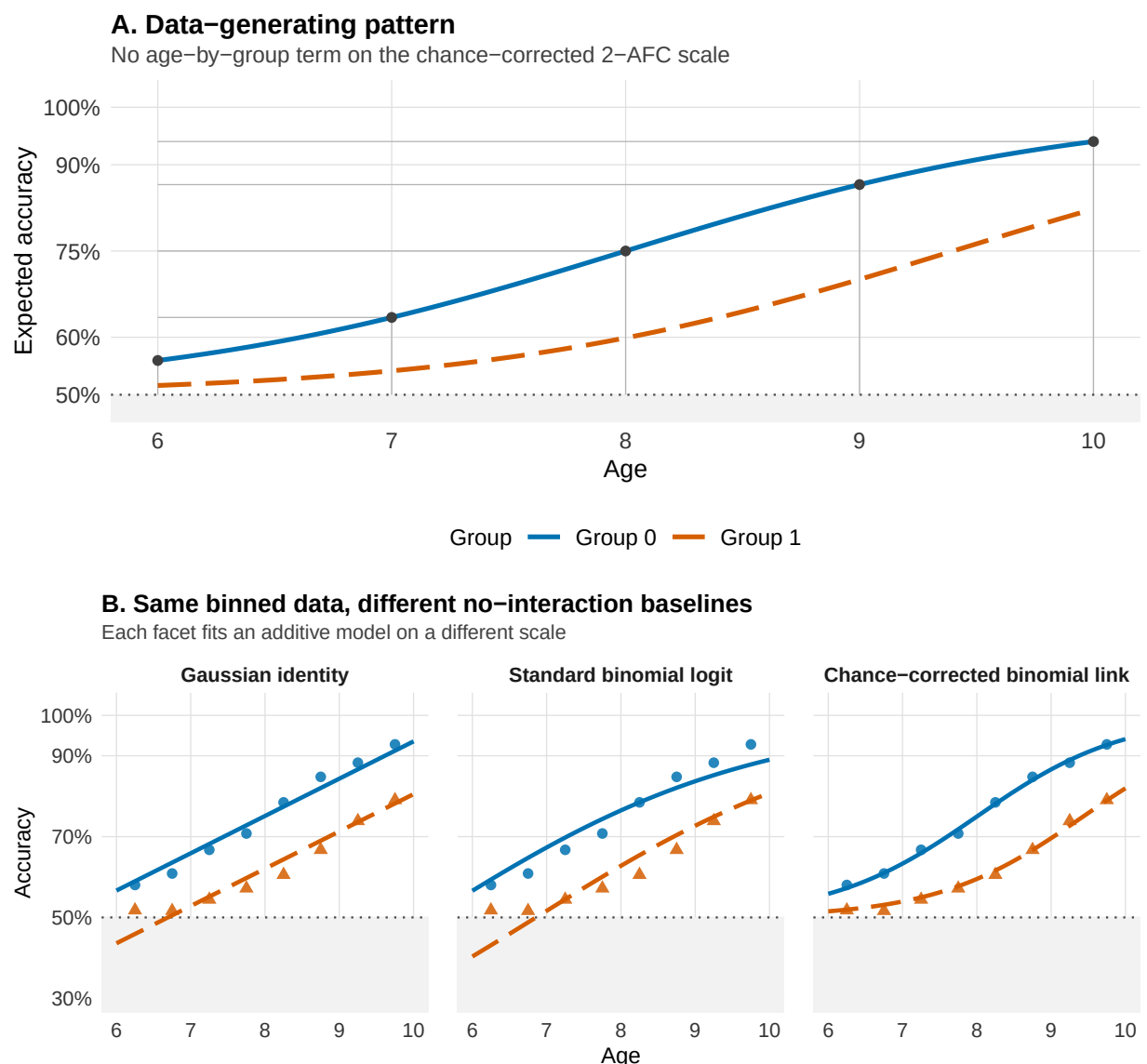


Figure 3

A simple count example. The two groups have the same age effect on the log-count scale and no age-by-group product term on that scale. After the inverse log link, expected error counts decrease nonlinearly on the observed scale, and absolute group differences change with age.

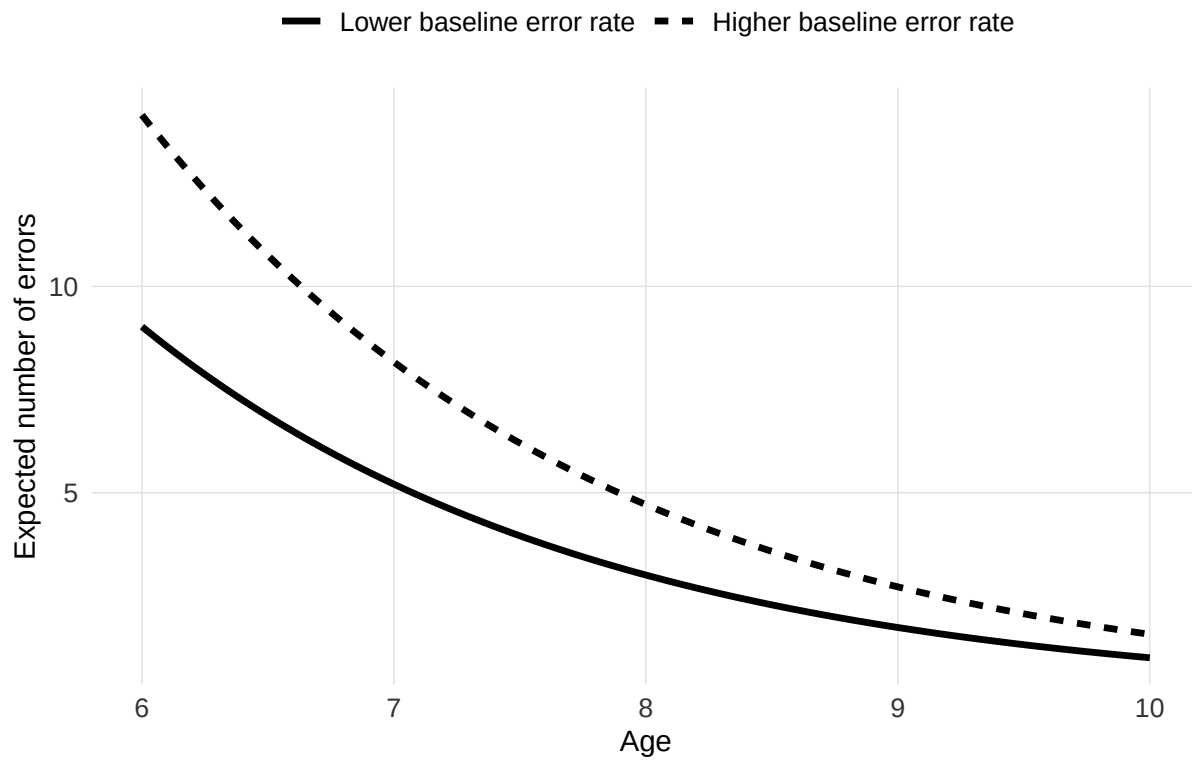


Figure 4

Fitted logit and probit models on one simulated binary dataset. The data were generated from a logistic curve, and both a binomial-logit and a binomial-probit GLM were fitted to the same responses. Panel A shows binned observed proportions together with the two fitted probability curves. Panel B shows the pointwise difference between the fitted probabilities, computed as logit minus probit. The curves can look nearly interchangeable while still implying a structured discrepancy across the predictor range. For interaction claims, this matters when the relevant comparison depends on regions where the two fitted links allocate different response-scale room for an effect.

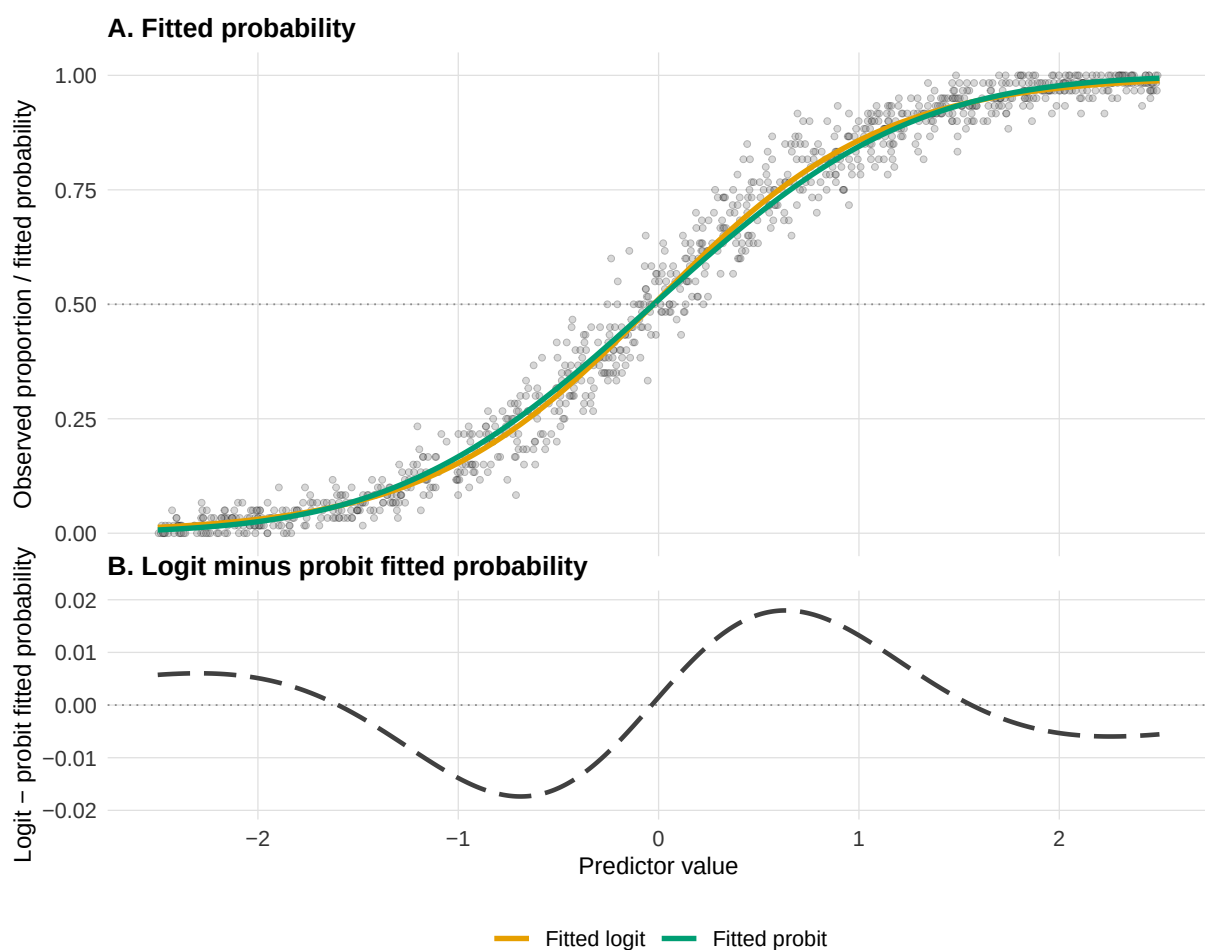


Figure 5

Simulation 1: forced-choice accuracy with a task-implied chance floor. Panel A shows the deterministic scenario curves generated by the chance-corrected logit model. The grey region marks values below the .50 chance floor, and the dashed horizontal line marks chance performance. Panel B shows the false-positive rate for the age-by-group product term across repeated simulations. Data are generated with no age-by-group interaction on the chance-corrected scale.

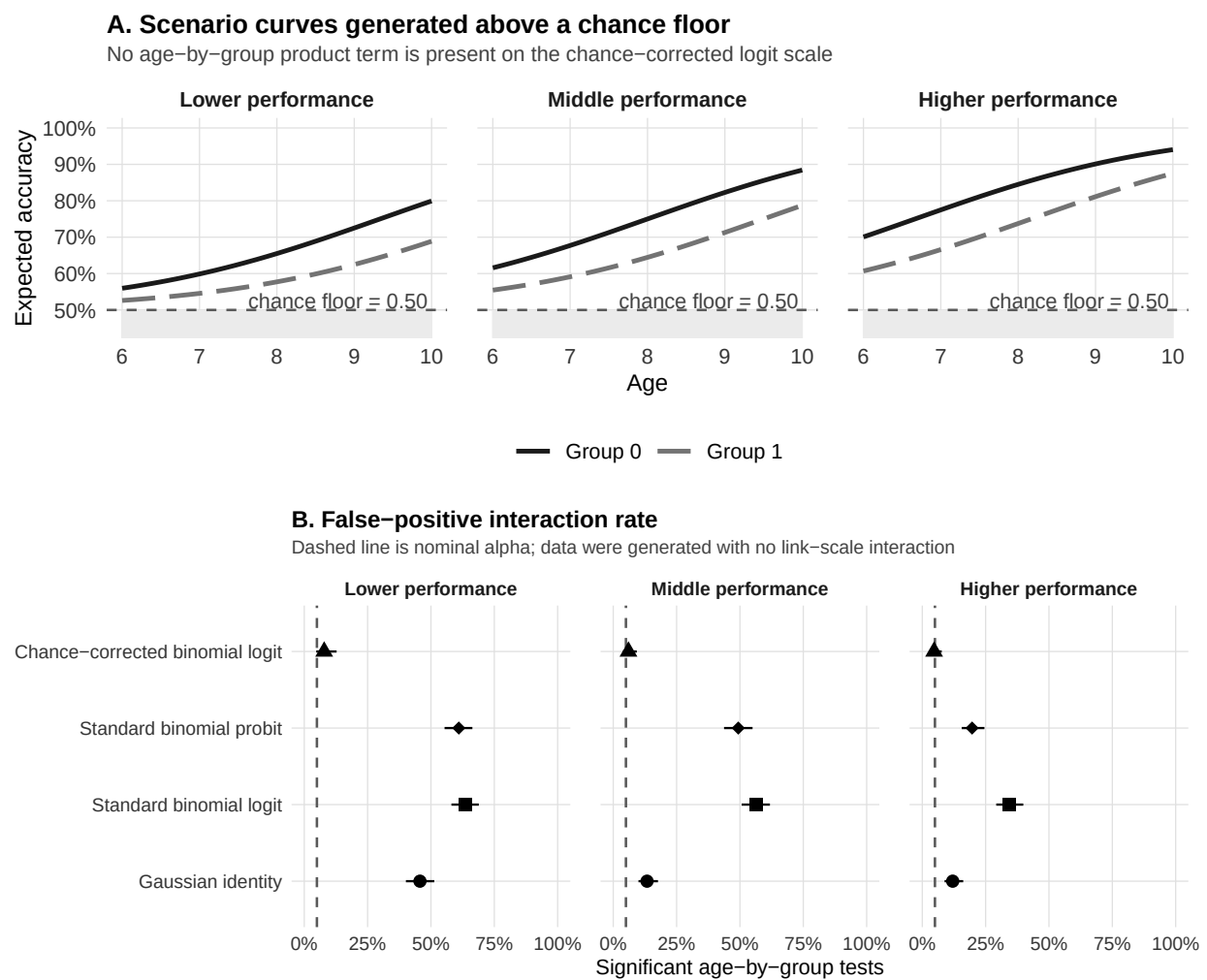


Figure 6

Simulation 2: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x -by-group product term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows the response-scale group gap implied by the same latent group difference; the gap can change across x because the observed score is compressed unevenly. Panel C shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because θ is known in the simulation, not a model available in ordinary applied data analysis.

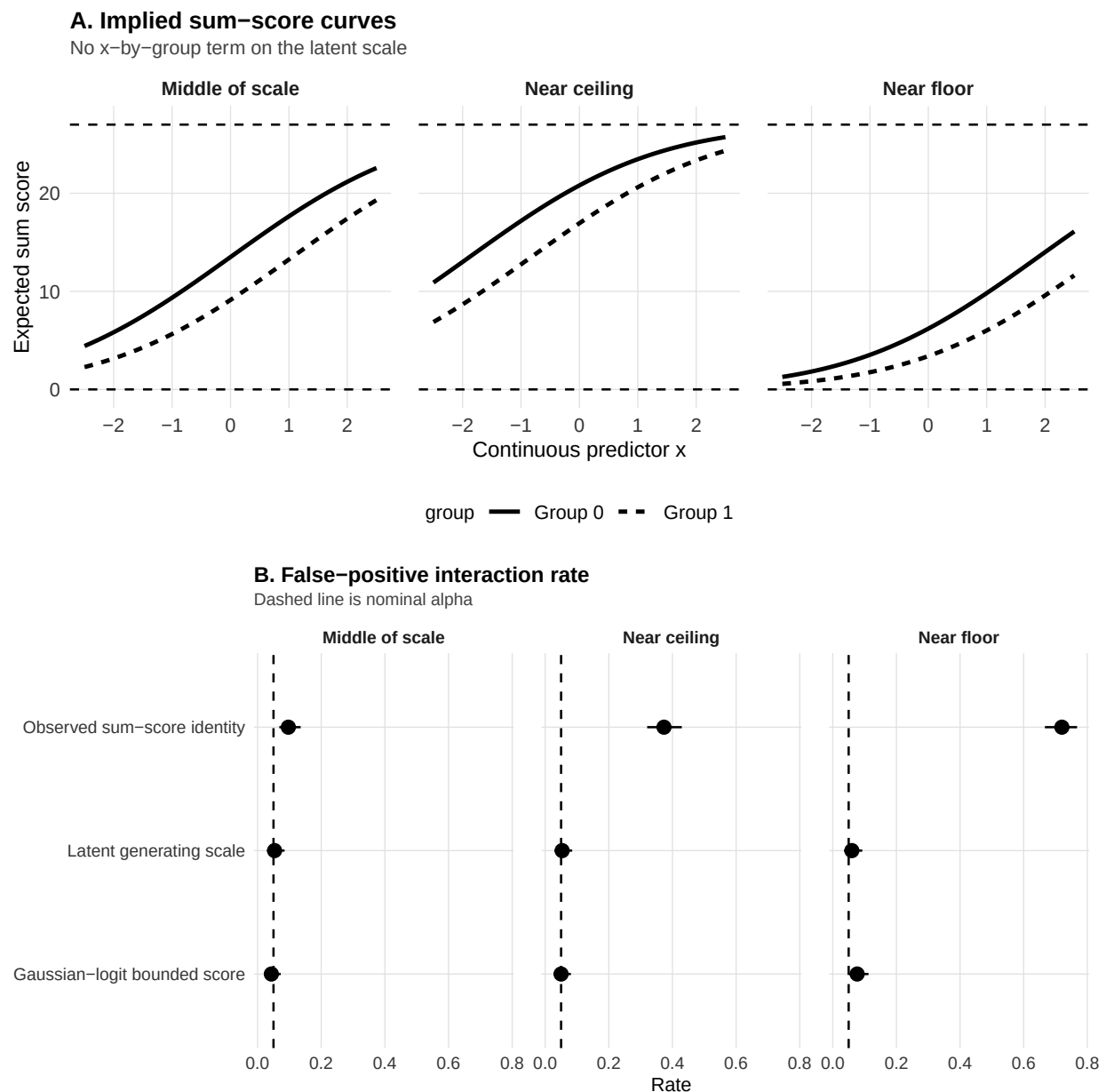


Figure 7

Simulation 3: within-family link choice in repeated-trial random-intercept GLMMs. Panel A shows the deterministic cell probabilities generated by additive logit or probit models at random intercept 0. Panel B shows the false-positive rate for the condition-by-group product term when the generating and fitted links are crossed. Error bars are Wilson 95% confidence intervals, and the dashed line marks nominal alpha.

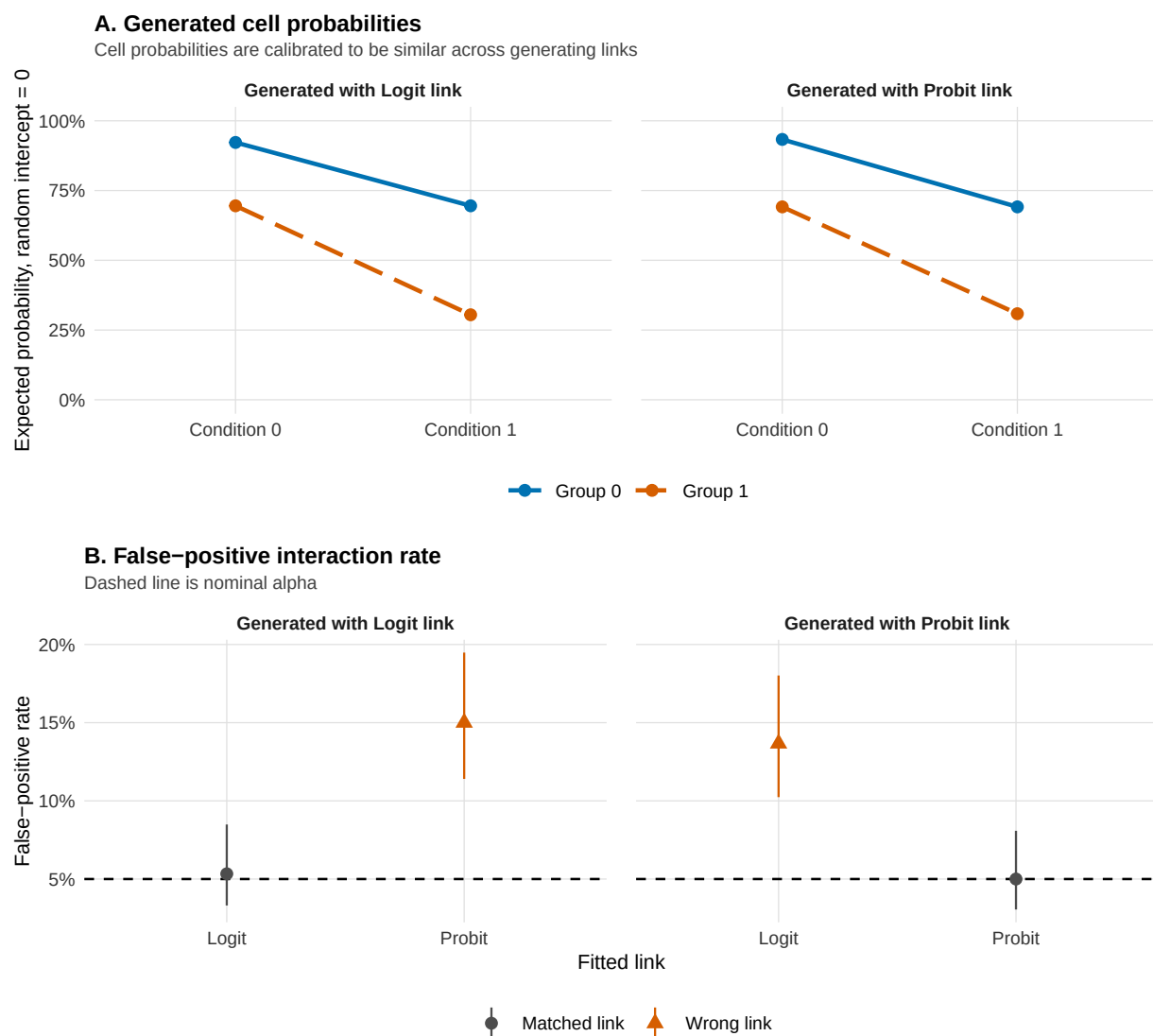


Figure 8

Diagnostic worked example. Panel A shows the deterministic generating pattern: a chance-corrected binomial model with a .50 floor and no age-by-group product term on the generating link scale. Panel B shows the false-positive rate for the age-by-group interaction after fitting the wrong standard binomial-logit link. Panel C shows diagnostic detection rates reported separately: DHARMa uniformity, DHARMa dispersion, DHARMa residual quantiles over fitted values, DHARMa residual quantiles over age, and a Pregibon-style added-term link check used as a secondary comparator. Error bars are Wilson 95% confidence intervals, and dashed lines mark nominal alpha.

